2018A F=ma Exam: Problem 22

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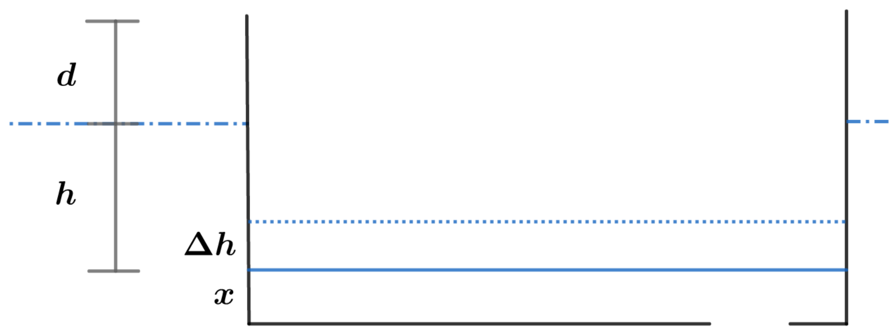
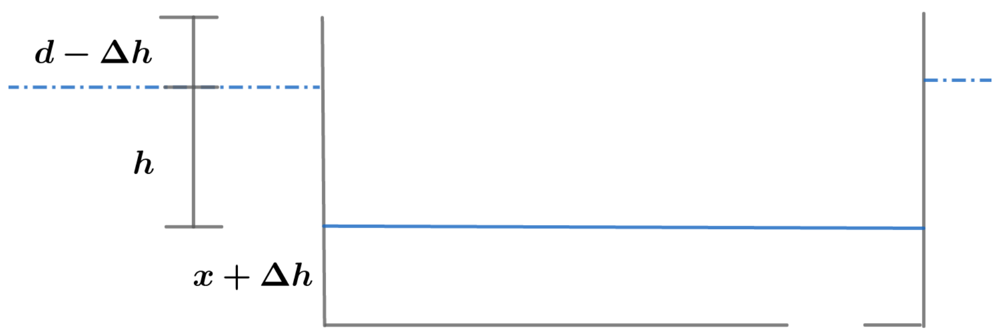


Suppose the boat is instantaneously at equilibrium and a small amount of water enters increasing the water level inside by Δh.



The boat will sink by Δh which increases the buoyant force by the weight of the additional water, restoring equilibrium. Note that the difference between the inner and outer water levels remains the same. Applying Bernoulli's equation to the inside and outside water,

$$P_0 + \frac{1}{2}\rho v^2 = P_0 + \rho g h$$

$$v = \sqrt{2gh}$$

we see the leak rate is constant. Thus, the answer is $\boxed{\text{A}}$.